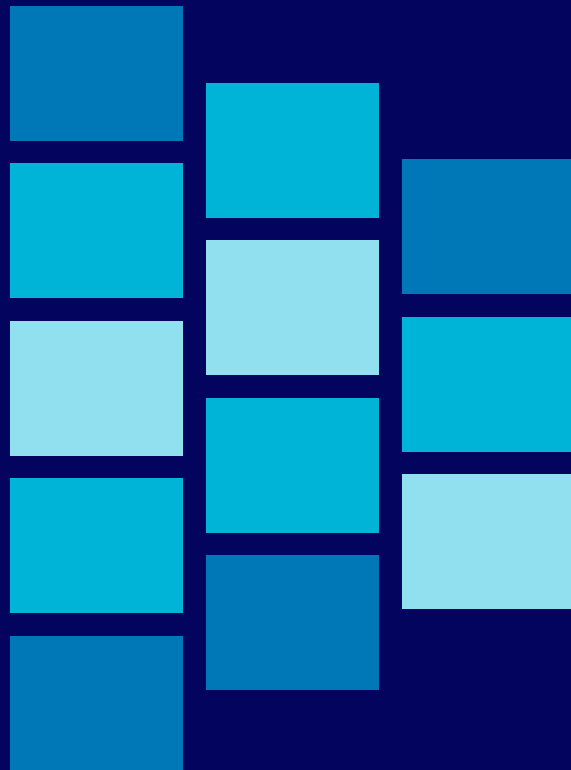


SmolVLM

Redefining Small and Efficient
Multimodal Models

Hugging Face & Stanford University

Compact Vision-Language Models for Edge Devices



256M–2.2B

Parameters

<1 GB

GPU RAM (256M)

80 tok/s

On-Device

13

Benchmarks

Large VLMs Are Powerful but Impractical for Edge Deployment

THE PROBLEM

- State-of-the-art VLMs require 20–80B parameters
- GPU memory: 15–40+ GB for inference
- Cloud dependency introduces latency and privacy risks
- Smartphones and edge devices: 4–8 GB RAM total

THE GAP

27.7 GB

Molmo-A1B-7B GPU RAM

80B

Idefics-80B Parameters

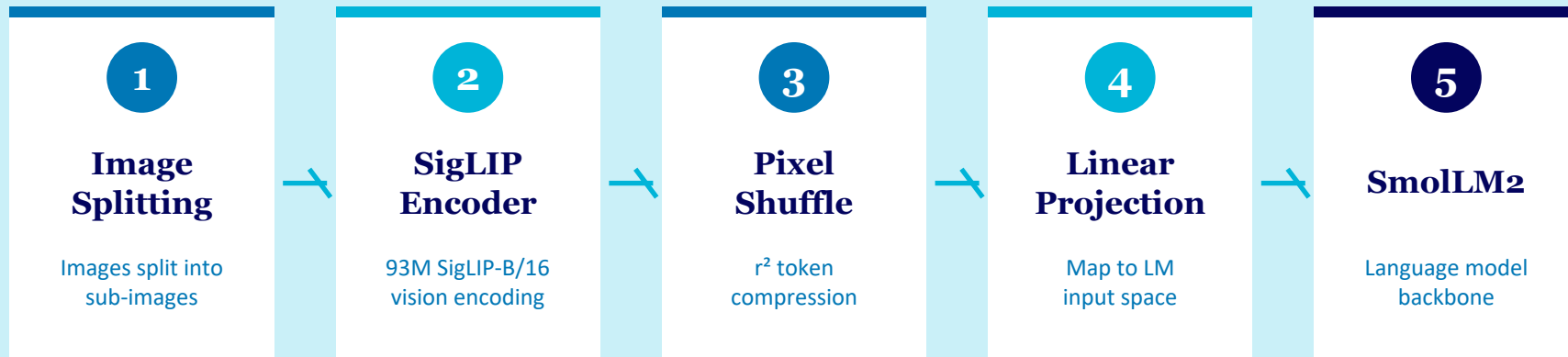
300×

Size Gap vs. SmolVLM-256M

Edge devices need VLMs that fit in <1 GB RAM — without sacrificing capability

SmolVLM Architecture

Image Splitting → Pixel Shuffle → SmolLM2 Backbone



Video: Frames are sampled and processed through the same pipeline

Balanced Encoder-LM Compute

Smaller encoder for compact LMs

Aggressive Token Compression

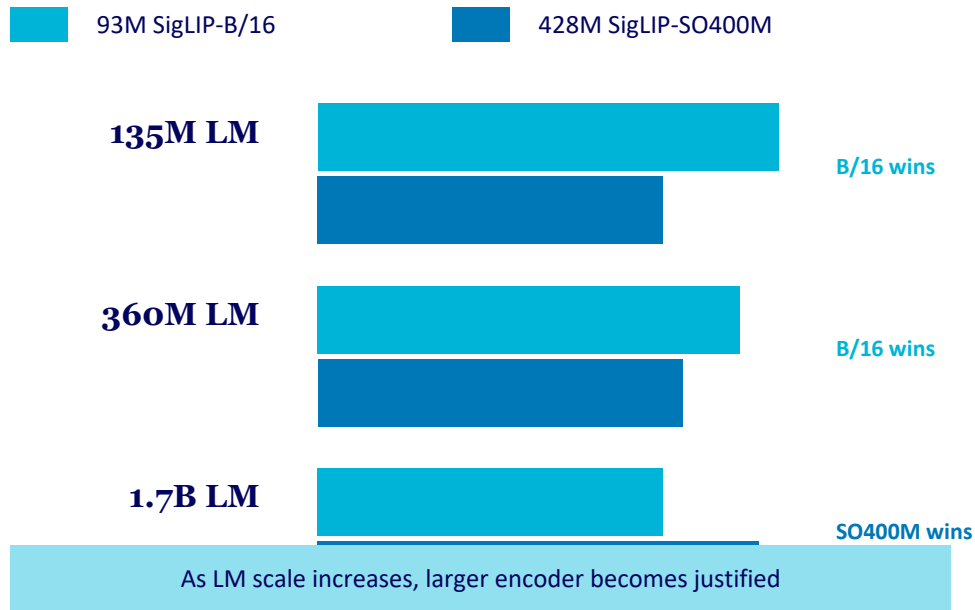
$r=4$ pixel shuffle for small models

Extended Context Length

2K → 16K via RoPE scaling

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1: Balanced encoder-LM parameter allocation



Key Finding

93M SigLIP-B/16 outperforms 428M SigLIP-SO400M when paired with the smallest LMs (135M)

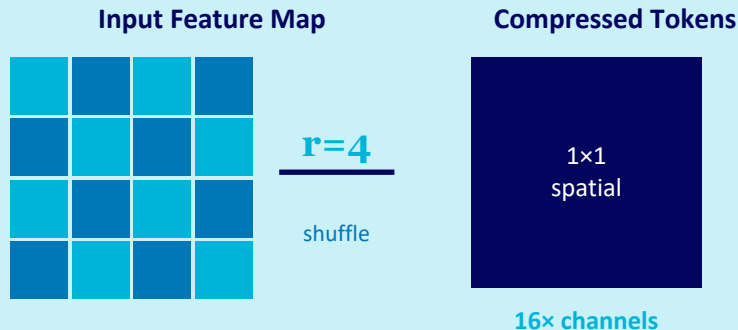
Crossover Point

Only at 1.7B LM scale does the larger encoder justify its 10% parameter overhead

Implication: Allocate compute to the LM, not the encoder, for compact VLMs

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Finding 3: Aggressive token compression for compact VLMs



16× token reduction at $r=4$ ($r^2 = 4^2 = 16$)

Pixel shuffle (space-to-depth) rearranges spatial features into channels:

- Spatial tokens reduced by r^2 ($r=4 \rightarrow 16\times$ fewer tokens)
- Channel depth increases to compensate
- Eases attention overhead for small models

Shuffle Factor Comparison

	$r=2$	$r=4$
Token Reduction	4×	16×
Best For	Large LMs	Small LMs
Attention Load	Moderate	Minimal
Long-Context	Standard	Improved

Small VLMs benefit from $r=4$ (16× reduction) while larger models prefer $r=2$ (4× reduction)

Visual Token Count

$r=2$: 4× tokens

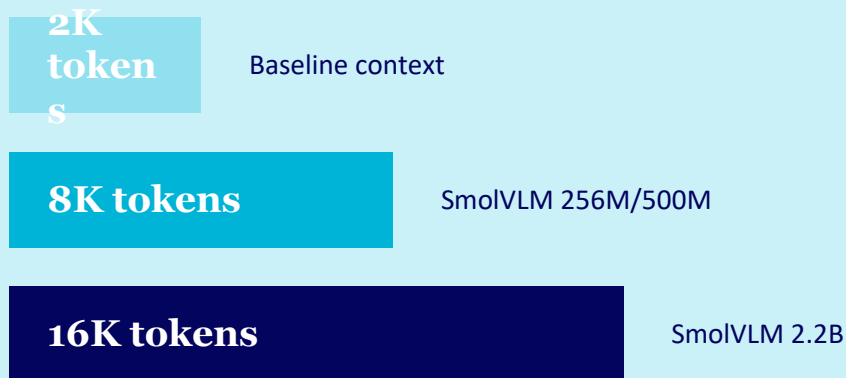
$r=4$

1/4 of $r=2$ tokens

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Finding 2: Context length scaling via RoPE base adjustment

Context Length Progression



RoPE Base Scaling

RoPE base: 10K → 273K

- Fine-tuning on mixed long/short-context data
- 2.2B model: 16K context length
- 256M / 500M: 8K context length

Why Context Length Matters

Higher Resolution Input

More tokens allow processing larger images at higher resolution without aggressive downsampling

Multi-Image Reasoning

Longer contexts support multiple images or video frames in a single forward pass

Detailed Document Understanding

Extended context enables processing full document pages with rich text and layout

Video Comprehension

More frames can be sampled and processed together, improving temporal understanding

8× context extension with minimal parameter overhead

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Category	Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7%
	AI2D	46.4%	59.2%	70.0%	71.0%
	ChartQA	55.6%	62.8%	68.7%	48.0%
	TextVQA	50.2%	60.2%	73.0%	61.5%
	DocVQA	58.3%	70.5%	80.0%	77.7%
	ScienceQA	73.8%	80.0%	89.6%	87.5%
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9%
	MathVista	35.9%	40.1%	51.5%	37.6%
	MMStar	34.6%	38.3%	46.0%	43.1%
Video	Video-MME	33.7%	42.2%	52.1%	45.0%
	MLVU	40.6%	47.3%	55.2%	48.2%
	Average	44.0%	51.0%	59.8%	—

0.8 GB

SmolVLM-256M

1.2 GB

SmolVLM-500M

4.9 GB

SmolVLM-2.2B

27.7 GB

Molmo-A1B-7B

SmolVLM-256M surpasses the 300× larger Idefics-80B | Baselines: Molmo-A1B-7B, InternVL2-2B

Real-Time Inference on iPhones

Up to 80 tokens/second on Apple Silicon via HuggingSnap

Throughput Benchmarks



Batch=64 throughput: SmolVLM-256M reaches ~1000 tok/s on high-end devices

Throughput scales roughly linearly with batch size for the smallest model

HuggingSnap iOS App

- Fully local inference — no cloud dependency
- Processes photos and videos on-device
- Privacy-preserving by design

0.8 GB

GPU RAM

SmolVLM-256M

Smartphones

batch=1

1.2 GB

GPU RAM

SmolVLM-500M

Tablets

batch=1

4.9 GB

GPU RAM

SmolVLM-2.2B

Laptops

batch=1

Key Takeaways

Design Principles for Efficient Multimodal Models

01

Balance Encoder-LM Compute

Smaller encoders (93M) outperform larger ones (428M) for compact LMs — allocate compute to the language model

02

Use Aggressive Token Compression

r=4 pixel shuffle delivers 16× token reduction for small models, easing attention overhead and improving long-context modeling

03

Extend Context Length

Scaling from 2K to 16K tokens via RoPE base adjustment unlocks higher resolution and multi-image reasoning

04

Sub-1GB Inference Is Achievable

SmolVLM-256M runs at 0.8 GB GPU RAM, outperforming the 300× larger LLaVA-80B

05

Fully Open-Sourced

All weights, datasets, training code, and HuggingSnap mobile app are open-sourced for reproducibility

Architectural efficiency matters more than raw scale.

SmolVLM demonstrates that strategic design choices — balanced compute, aggressive compression, and extended context — enable competitive multimodal performance at a fraction of the resource cost.

[arXiv: 2504.05299](#)

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